The shift left strategy

The 'shift left' strategy is a popular topic in product development circles, frequently discussed at conferences and industry events. Its core principle is to start the development process with a comprehensive understanding of the product. By doing so, we can avoid costly and time-consuming rework later in the development cycle. Implementing the shift left strategy means proactively addressing potential issues and refining the product concept from the outset. This approach helps streamline the development process, reduce costs, and accelerate time to market.

cal testing commences. This method is not intended to eliminate physical testing but rather to complement it by minimising the number of required tests. For example, simulations can be employed to evaluate a motor's performance under various conditions or to assess the thermal management of a battery. By implementing these simulations early in the design phase, potential issues can be identified and addressed before moving on to physical prototypes.

The validation process encompasses multiple levels, ranging from individual components to entire systems. At the component level, simulations might concentrate on specific aspects such as motor performance or battery thermal management. Subsystem-level validation could involve examining how a battery management system (BMS) reacts to different scenarios, such as disengaging a cell due to a low state of charge or managing fire hazards. Ultimately, vehicle-level validation assesses the overall performance under various conditions, including different terrains and climates.

One of the challenges in virtual validation is ensuring that simulations accurately reflect real-world behaviour. This necessitates not only the development of detailed models but also their calibration to accurately capture the underlying physics. For instance, a battery model should replicate the temperature changes during charge and discharge cycles as they would occur in an actual battery.

As we progress towards more comprehensive vehicle-level simulations, the environment in which the vehicle operates becomes a critical factor. The performance of a vehicle can differ significantly depending on whether it is driven in a city, on a highway, or in mountainous regions. Integrating environmental factors into simulations aids in making more accurate predictions about vehicle behaviour in different conditions.

Supporting the advancement of virtual validation is reduced order modelling, a technique that compresses complex physics-based models, making them more suitable for integration into system-level simulations that run closer to real-time. Additionally, machine learning plays a crucial role in calibrating models based on real-world data, ensuring that they accurately reflect the variability and uncertainties of actual operating conditions.

Improving cars with digital twins

A critical aspect that original equipment manufacturers (OEMs) focus on is the integration of numerous sensors within vehicles to capture a wealth of telemetry data. This data holds the key to unlocking a deeper understanding of vehicle performance and behaviour. As we move towards personalisation and customisation of vehicles, the concept of over-the-air (OTA) updates emerges as a pivotal tool for enhancing vehicle performance. But how can we achieve this? The answer lies in the utilisation of digital twins and the Internet of Things (IoT).

The core idea is to create a virtual replica of the vehicle in a cloud or IoT environment. This digital twin continuously learns from the real-world data collected by the vehicle's sensors and adjusts itself accordingly. By doing so, it mirrors the behaviour of the actual vehicle based on the driver's real-world driving patterns. This capability allows for the prediction of future performance, such as estimating the battery's capacity loss after say 10,000 kilometres or identifying potential failures.

Moreover, the digital twin enables the customisation of software controls to optimise performance for specific driving patterns. This is where OTA updates come into play, allowing for the personalisation of the vehicle's software to enhance its performance based on individual driving habits.

There are use cases in the open domain where we have demonstrated how physics-based models can be integrated into a cloud-based environment, specifically focusing on software-defined vehicles. This integration showcases the potential of digital twins and IoT in revolutionising vehicle performance and personalisation, ultimately providing a more tailored and efficient driving experience.

The automotive industry is experiencing a rapid transformation in product development. The shift towards electric vehicles, coupled with the increasing complexity of vehicle systems, requires innovative approaches to streamline the development process. Virtualisation, MBSE, and virtual validation are key strategies being adopted to address these challenges. By embracing these technologies, companies can accelerate product development, reduce costs, and deliver vehicles that meet the demands of the modern market. EFY

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